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CUET UG

General Aptitude Test

Quantitative Reasoning

Section III · General Test · Complete Theory, Visual Frameworks & Exam Mastery

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Syllabus Alignment & Chapter Architecture

Sub-Topic Domain	Official NTA CUET UG Section III Syllabus Scope	High-Frequency Cognitive Skills Tested
Number Theory & HCF-LCM	Divisibility laws, prime factorization, properties of numbers, remainders, LCM and HCF application models.	Multi-digit divisibility shortcuts, cyclic unit digits, remainder theorem, and co-prime factorization.
Commercial Arithmetic	Percentages, profit and loss, successive discounts, marked price, simple and compound interest, partnerships.	Multiplier speed calculations, dishonest dealer trick problems, 2-year/3-year CI-SI difference formulas.
Rates & Kinematics	Time, speed, distance, relative velocity, trains crossing platforms/poles, boats and streams, circular tracks.	Conversion factors ($\frac{5}{18}$, $\frac{18}{5}$), upstream/downstream symmetry, harmonic mean applications.
Work & Cistern Dynamics	Time and work unitary methods, efficiency proportions, work and wages, negative work in cistern leaks.	LCM total work allocation, alternating day operations, combined efficiency computation.
Algebra & Progressions	Linear and quadratic equations, discriminant analysis, algebraic identities, AP, GP, AM-GM inequalities.	Symmetric rational substitutions ($x + \frac{1}{x}$), infinite geometric series, roots relationship ($b^2 - 4ac$).
Geometry & Coordinate	Angles, triangles (Pythagorean triplets, similarity), circles, tangents, Cartesian coordinate geometry.	Standard triplets, cyclic quadrilaterals, chord properties, distance formula, slope perpendicularity.
Mensuration 2D & 3D	Plane figures (triangles, polygons, circles), solids (cubes, cylinders, cones, spheres), recasting of solids.	Heron's formula, CSA vs TSA distinction, volume conservation during melting and recasting.

1. Number Systems & Divisibility Mechanics

Quantitative Reasoning in CUET UG evaluates computational agility, mathematical intuition, and algebraic precision rather than brute-force mechanical arithmetic. Master candidates eliminate options through divisibility tests, digital roots, and unit digits without computing multi-digit multiplications.

Classification of Numbers

The real number system $\mathbb{R}$ divides into Rational numbers $\mathbb{Q}$ (expressible as $\frac{p}{q}$ where $p, q \in \mathbb{Z}, q \neq 0$) and Irrational numbers (non-terminating, non-repeating decimals such as $\sqrt{2}, \sqrt{3}, \pi, e$).

- **Prime Numbers:** Integers strictly greater than 1 having exactly two positive divisors (1 and itself). The only even prime is 2. Primes up to 50: $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47$ (exactly 15 primes). Primes between 50 and 100: $53, 59, 61, 67, 71, 73, 79, 83, 89, 97$ (10 primes, total 25 primes under 100).

- **Coprime Integers:** Two integers a and b are coprime if and only if $\text{HCF}(a, b) = 1$. They do not need to be individual prime numbers (e.g., 8 and 9 are coprime).
- **Twin Primes:** Pairs of prime numbers differing by exactly 2 (e.g., $(3, 5), (5, 7), (11, 13), (17, 19), (29, 31)$).

Multi-Digit Divisibility Rules & Shortcuts

- **Divisibility by 2^n:** An integer is divisible by $2, 4, 8, 16$ if its last $1, 2, 3, 4$ digits respectively form a number divisible by $2, 4, 8, 16$.
- **Divisibility by 3 and 9:** A number is divisible by 3 (or 9) if the digital sum of all its digits is divisible by 3 (or 9).
- **Divisibility by 11:** The difference between the sum of digits in odd places and the sum of digits in even places (counted from right) must be either 0 or a multiple of 11.
- **Divisibility by 7, 11, 13 (Group of 3 Rule):** Form 3-digit groups from right to left. The alternating difference between odd groups and even groups must be divisible by 7, 11, or 13.
  - **Example:** For $4,832,198$: Groups are $198, 832, 4$. Alternating sum: $(198 + 4) - 832 = 202 - 832 = -630$. Since $\frac{630}{7} = 90$, the original number is strictly divisible by 7.
- **Composite Divisibility Law:** If a number is divisible by coprime factors p and q, it is divisible by their product $p \times q$.
  - Divisible by $72 \iff$ Divisible by both 8 and 9 (since $\text{HCF}(8, 9) = 1$).
  - Divisible by $88 \iff$ Divisible by both 8 and 11.
  - Divisible by $99 \iff$ Divisible by both 9 and 11.

Cyclicity of Unit Digits

The last digit of a^n repeats in periodic cycles of length at most 4:

- **Cyclicity 1:** Numbers ending in $0, 1, 5, 6$ always retain their unit digit for any positive integer power (5^n ends in 5, 6^n ends in 6).
- **Cyclicity 2:** Numbers ending in 4 and 9:
  - $4^{\text{odd}} \to 4$, $4^{\text{even}} \to 6$.
  - $9^{\text{odd}} \to 9$, $9^{\text{even}} \to 1$.
- **Cyclicity 4:** Numbers ending in $2, 3, 7, 8$:
  - Divide the exponent by 4. If remainder is $r \in \{1, 2, 3\}$, unit digit is a^r.
  - If remainder is 0, the effective exponent is 4 (a^4).
  - **Example:** Find unit digit of 7^{135}. Here $135 \mod 4 = 3$. Unit digit $= 7^3 = 343 \to *3*$.

Remainder Theorems & Power Reductions

- **Fermat's Little Theorem:** If p is a prime number and $\text{HCF}(a, p) = 1$, then:
$$a^{p-1} \equiv 1 \pmod{p}$$
- **Wilson's Theorem:** If p is prime, then $(p-1)! \equiv -1 \equiv (p-1) \pmod{p}$.
- **Successive Remainder Model:** When number N is divided successively by divisors $d_1, d_2, \ldots$ giving remainders $r_1, r_2, \ldots$, work backwards from the final quotient q: $N = d_1[d_2 q + r_2] + r_1$.

2. Factors, Multiples, HCF & LCM Architecture

Let $N = p_1^a \times p_2^b \times p_3^c$ be the canonical prime factorization of natural number N.

- **Total Number of Factors:** $N_f = (a+1)(b+1)(c+1)$.
- **Number of Even Factors:** If $p_1 = 2$, $N_{\text{even}} = a(b+1)(c+1)$.
- **Number of Odd Factors:** $N_{\text{odd}} = (b+1)(c+1)$.

- **Sum of Factors:** $S_f = \left(\frac{p_1^{a+1}-1}{p_1-1}\right) \times \left(\frac{p_2^{b+1}-1}{p_2-1}\right) \times \left(\frac{p_3^{c+1}-1}{p_3-1}\right)$.
- **Fundamental HCF-LCM Identity:** For any two positive integers A and B:

$$\text{HCF}(A, B) \times \text{LCM}(A, B) = A \times B$$

- **Fractions Rule:**

$$\text{HCF}\left(\frac{a}{b}, \frac{c}{d}\right) = \frac{\text{HCF}(a, c)}{\text{LCM}(b, d)}, \quad \text{LCM}\left(\frac{a}{b}, \frac{c}{d}\right) = \frac{\text{LCM}(a, c)}{\text{HCF}(b, d)}$$

Standard Exam Models for HCF & LCM

Question Archetype	Mathematical Formulation	Operational Solution Method
Greatest divisor leaving equal remainder	Find largest x dividing a, b, c leaving the same remainder r in each case.	$x = \text{HCF}(\lvert a-b\rvert, \lvert b-c\rvert, \lvert c-a\rvert)$. The remainder r is not required upfront.
Smallest number leaving constant remainder	Find smallest N which when divided by a, b, c leaves remainder R in each case.	$N = [\text{LCM}(a, b, c) \times k] + R$, where $k = 1$ for the smallest natural value.
Smallest number leaving variable remainders	Dividing by a, b, c leaves remainders r_1, r_2, r_3 where $(a - r_1) = (b - r_2) = (c - r_3) = d$.	$N = [\text{LCM}(a, b, c) \times k] - d$. The constant deficit d is subtracted.

3. Commercial Arithmetic: Percentages, Profit, Loss & Discount

The Decimal Multiplier Technique

Avoid setting up traditional $\frac{x}{100}$ equations. Use direct fractional or decimal multipliers:

- A 25% increase corresponds to multiplying by $1 + 0.25 = 1.25$ (or $\frac{5}{4}$).
- A 16.67% decrease corresponds to multiplying by $1 - \frac{1}{6} = \frac{5}{6}$.
- **Fraction Equivalents to Memorize:**

$$\frac{1}{2} = 50\%,\ \frac{1}{3} = 33.33\%,\ \frac{1}{4} = 25\%,\ \frac{1}{5} = 20\%,\ \frac{1}{6} = 16.67\%,\ \frac{1}{7} = 14.28\%,\ \frac{1}{8} = 12.5\%,\ \frac{1}{9} = 11.11\%,\ \frac{1}{12} = 8.33\%$$

Successive Percentage Changes & Product Constancy

- When an entity changes successively by $+a\%$ and $+b\%$, the net percentage change is:

$$\text{Net Percentage Change} = a + b + \frac{a \times b}{100}\%$$

(Note: Apply negative signs for discounts or percentage reductions).

- If a value increases by $x\%$ and subsequently decreases by $x\%$, the overall outcome is ALWAYS a net decrease of:

$$\text{Net Loss} = -\left(\frac{x}{10}\right)^2\% = -\left(\frac{x^2}{100}\right)\%$$

- **Product Constancy Law:** If Price × Consumption = Expenditure, and Price increases by $\frac{1}{x}$, Consumption must decrease by $\frac{1}{x+1}$ to keep expenditure strictly constant.
  - **Example:** If price of sugar rises by 25% ($\frac{1}{4}$), consumption must fall by $\frac{1}{4+1} = \frac{1}{5} = 20\%$.

Commercial Trading: Profit, Loss & Marked Price Ladder

THE COST-PRICE TO MARKED-PRICE PIPELINE

The commercial trading cycle proceeds in three distinct pricing tiers:

$$\text{Cost Price (CP)} \overset{\text{Markup \%}}{\rightarrow} \text{Marked Price (MP)} \overset{\text{Discount \%}}{\rightarrow} \text{Selling Price (SP)}$$

- Profit is ALWAYS measured against Cost Price: $\text{Profit \%} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100$.
- Discount is ALWAYS measured against Marked Price: $\text{Discount \%} = \frac{\text{MP} - \text{SP}}{\text{MP}} \times 100$.
- Relation connecting Markup and Discount:

$$\text{SP} = \text{CP}\left(1 + \frac{\text{Profit \%}}{100}\right) = \text{MP}\left(1 - \frac{\text{Discount \%}}{100}\right)$$

- **Dishonest Trader Formula:** When a merchant claims to sell at Cost Price but uses a false weight of w grams instead of standard W grams:

$$\text{Gain \%} = \frac{\text{Error}}{\text{True Value} - \text{Error}} \times 100 = \frac{W - w}{w} \times 100$$

- **Combined Deception:** If merchant marks up goods by $m\%$ AND uses false weight w instead of W:

$$\text{Overall Multiplier} = \left(1 + \frac{m}{100}\right) \times \left(\frac{W}{w}\right) \rightarrow \text{Net Profit \%} = (\text{Overall Multiplier} - 1) \times 100$$

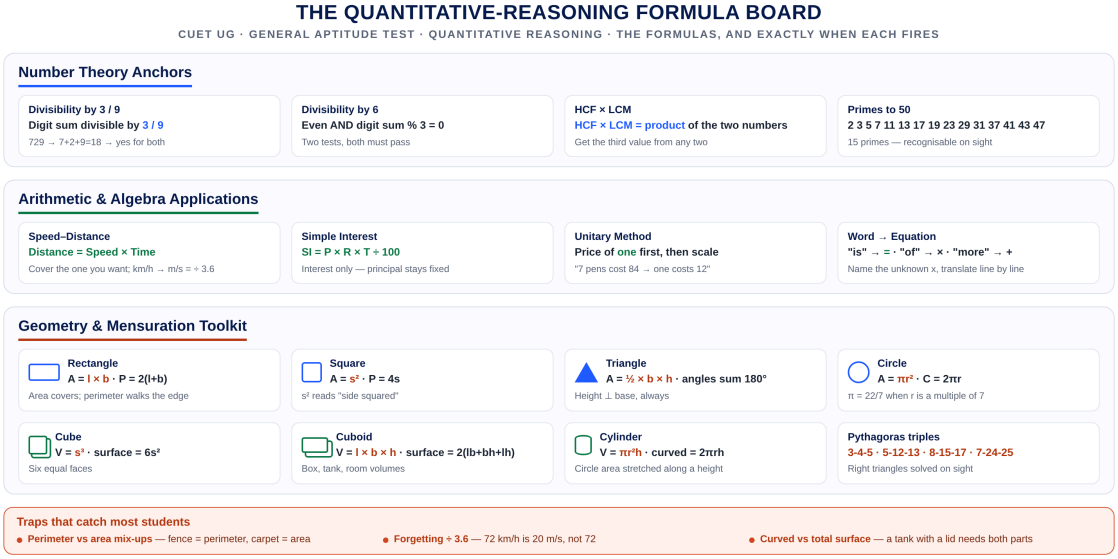


Figure 1: The Master Quantitative Reasoning Formula Board: Architectural visual summary of operational algebraic identities, plane mensuration formulas, and commercial calculation paths.

4. Financial Mathematics: Simple & Compound Interest

Simple Interest (SI)

Simple Interest accumulates linearly only on the initial principal P:

$$\text{SI} = \frac{P \times R \times T}{100}, \quad \text{Amount } A = P + \text{SI} = P\left(1 + \frac{RT}{100}\right)$$

- If a sum of money doubles itself in T years at Simple Interest, the rate of interest is $R = \frac{100}{T}\%$.
- If a sum becomes n times itself in T years at Simple Interest: $T = \frac{(n-1)\times 100}{R}$.
- **Installments in SI:** If a debt of Rs A is paid in n equal annual installments of Rs x at $R\%$ per annum:

$$A = nx + \frac{xR}{100} \times \frac{n(n-1)}{2}$$

Compound Interest (CI) & Compounding Frequencies

Compound Interest reinvests earned interest, generating exponential growth:

$$A = P\left(1 + \frac{R}{100}\right)^n, \quad \text{CI} = A - P = P\left[\left(1 + \frac{R}{100}\right)^n - 1\right]$$

- **Semi-Annual Compounding:** Rate becomes $\frac{R}{2}\%$ per half-year; time periods become $2n$.
- **Quarterly Compounding:** Rate becomes $\frac{R}{4}\%$ per quarter; time periods become $4n$.

Critical Difference Formulas (CI vs SI)

CUET high-frequency questions target the monetary difference between Compound and Simple Interest over identical principal P and annual rate $R\%$:

- **For 2 Years:**

$$\text{Difference } (D_2) = \text{CI}_2 - \text{SI}_2 = P\left(\frac{R}{100}\right)^2$$

- **For 3 Years:**

$$\text{Difference } (D_3) = \text{CI}_3 - \text{SI}_3 = P\left(\frac{R}{100}\right)^2 \times \left(3 + \frac{R}{100}\right) = D_2 \times \left(3 + \frac{R}{100}\right)$$

- **Rule of 72:** To find the approximate number of years required for an investment to double at annual compound rate $R\%$, compute:

$$T_{\text{double}} \approx \frac{72}{R}$$

5. Ratios, Mixtures & The Alligation Cross-Rule

Ratios, Proportions & Partnerships

- **Proportion:** $a : b :: c : d \iff ad = bc$ (Product of extremes equals product of means).
- **Mean Proportional:** Between a and b is $\sqrt{ab}$. Third proportional to a, b is $\frac{b^2}{a}$.
- **Partnership Capital Allocation:** Profit earned by partners is distributed directly in proportion to the product of their invested capital and the duration of investment:

$$\text{Profit Ratio} = (C_1 \times T_1) : (C_2 \times T_2) : (C_3 \times T_3)$$

If a working partner receives an active management fee (salary/commission), it is deducted from the gross profit first, and the remaining net profit is shared across all partners.

The Alligation Cross-Model

The Alligation method is an operational shortcut to determine the ratio in which two ingredients of prices C_1 (cheaper) and C_2 (dearer) must be blended to produce a mixture of mean price M:

$$\frac{\text{Quantity of Cheaper}}{\text{Quantity of Dearer}} = \frac{C_2 - M}{M - C_1}$$

Successive Dilution / Repeated Removal Formula

If a container initially holds V litres of pure liquid (e.g., milk or wine), and x litres are withdrawn and replaced with water n times successively, the quantity of pure liquid remaining is:

$$Q_{\text{remaining}} = V \times \left(1 - \frac{x}{V}\right)^n$$

The quantity of water present in the final mixture equals $V - Q_{\text{remaining}}$.

6. Kinematics: Speed, Distance, Trains & Watercraft

Fundamentals & Unit Standardization

$$\text{Distance} = \text{Speed} \times \text{Time}, \quad 1 \text{ km/h} = \frac{5}{18} \text{ m/s}, \quad 1 \text{ m/s} = \frac{18}{5} \text{ km/h}$$

- **Average Speed:** Defined strictly as Total Distance divided by Total Time:
  - If a journey covers equal distance d at speed u and returns at speed v:

$$\text{Average Speed} = \frac{2uv}{u+v} \quad \text{(Harmonic Mean)}$$

  - If a body travels for equal time t at speed u and speed v:

$$\text{Average Speed} = \frac{u+v}{2} \quad \text{(Arithmetic Mean)}$$

Train Dynamics & Extended Objects

- **Crossing a Negligible Object (Pole, Standing Person, Tree, Signal Post):**

$$\text{Distance Traveled} = L_{\text{train}}, \quad \text{Time} = \frac{L_{\text{train}}}{S_{\text{train}}}$$

- **Crossing an Extended Object (Platform, Bridge, Tunnel, Stationary Train):**

$$\text{Distance Traveled} = L_{\text{train}} + L_{\text{object}}, \quad \text{Time} = \frac{L_{\text{train}} + L_{\text{object}}}{S_{\text{train}}}$$

- **Relative Velocity of Two Moving Trains:**
  - Moving in Opposite Directions: $S_{\text{relative}} = S_1 + S_2$.
  - Moving in Same Direction: $S_{\text{relative}} = |S_1 - S_2|$.
  - Time taken to cross each other:

$$\text{Time} = \frac{L_1 + L_2}{S_{\text{relative}}}$$

Boats, Streams & Circular Races

Let the speed of the boat in still water be u km/h, and the speed of the water current be v km/h ($u > v$):

- **Downstream Speed (D):** The stream accelerates the boat: $D = u + v$.
- **Upstream Speed (U):** The stream retards the boat: $U = u - v$.
- **Speed of Boat in Still Water:** $u = \frac{D+U}{2}$.
- **Speed of Current / Stream:** $v = \frac{D-U}{2}$.

- **Circular Track Races:** Two runners starting at the same point with speeds u and v on track of circumference L:
  - Time to meet for first time anywhere: $T = \frac{L}{u-v}$ (same direction); $T = \frac{L}{u+v}$ (opposite direction).
  - Time to meet at the starting point: $T = \text{LCM}\left(\frac{L}{u}, \frac{L}{v}\right)$.

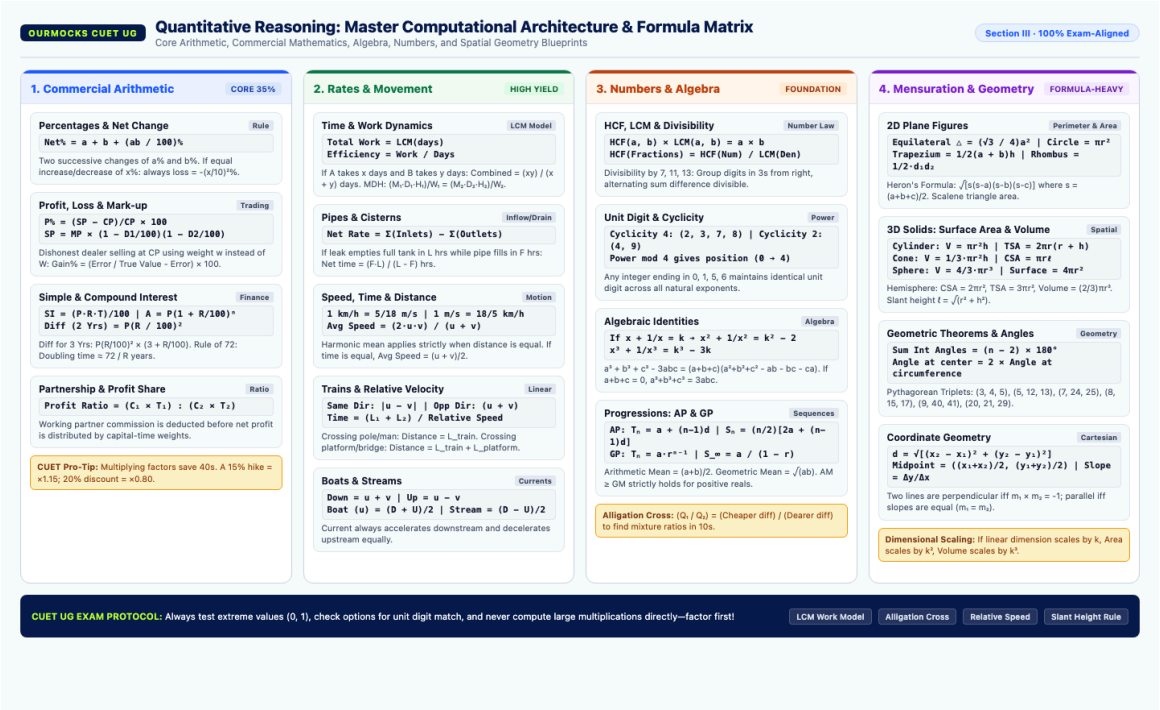


Figure 2: The Master Architecture of Quantitative Reasoning: Four foundational pillars mapping commercial arithmetic, motion and rates, algebraic structures, and spatial geometry mechanics.

7. Work, Wages & Fluid Flow Dynamics

The Total Work LCM Methodology

Rather than assuming total work to be 1 and dealing with cumbersome fractions, assume Total Work to be the **LCM of the individual days taken**.

- **Operational Workflow:**
  1. Calculate Total Work = $\text{LCM}(d_1, d_2, \ldots)$.
  2. Determine individual daily efficiency: Efficiency = $\frac{\text{Total Work}}{\text{Days}}$.
  3. Combine efficiencies: Total time taken equals Total Work divided by Combined Efficiency.
- **Two-Person Reciprocal Rule:** If A takes x days and B takes y days, together they take:

$$T_{\text{combined}} = \frac{xy}{x+y} \text{ days}$$

- **The Master MDH Manpower Equation:**

$$\frac{M_1 \times D_1 \times H_1 \times E_1}{W_1} = \frac{M_2 \times D_2 \times H_2 \times E_2}{W_2}$$

  Where M is workers, D is days, H is daily hours, E is efficiency, and W is units of work produced (or wages earned).
- **Alternating Work Days:** If A and B work on alternate days starting with A:
  - Form a two-day cycle unit: Work in 2 days = $E_A + E_B$.
  - Number of full cycles = $\left\lfloor \frac{\text{Total Work}}{E_A + E_B} \right\rfloor$.

- Compute remaining fractional work for the opening worker.

Pipes, Cisterns & Negative Work

Inflow pipes perform **positive work** (filling the tank), whereas drainage pipes or leaks perform **negative work** (emptying the tank).

- If Pipe A fills a tank in F hours and Leak B empties the full tank in L hours ($L > F$):

$$\text{Net Hourly Work} = \frac{1}{F} - \frac{1}{L} = \frac{L-F}{FL} \rightarrow \text{Time to fill} = \frac{F \times L}{L-F} \text{ hours}$$

- When multiple inlets and outlets operate simultaneously:

$$\text{Net Work Rate} = \sum(\text{Inlet Rates}) - \sum(\text{Outlet Rates})$$

8. Elementary Algebra, Identities & Progressions

Essential Algebraic Identities

CUET tests rapid algebraic simplification using symmetric identities:

- $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$
- $(a+b)^2 - (a-b)^2 = 4ab$
- $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
- $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$
  - **Golden Identity Corollary:** If $a+b+c=0$, then $a^3+b^3+c^3 = 3abc$.

Symmetric Rational Inverses ($x + \frac{1}{x}$)

If $x + \frac{1}{x} = k$:

- $x^2 + \frac{1}{x^2} = k^2 - 2$
- $x^3 + \frac{1}{x^3} = k^3 - 3k$
- $x^4 + \frac{1}{x^4} = (k^2-2)^2 - 2$
- $x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x}) = (k^2-2)(k^3-3k) - k$

If $x - \frac{1}{x} = m$:

- $x^2 + \frac{1}{x^2} = m^2 + 2$
- $x^3 - \frac{1}{x^3} = m^3 + 3m$

Quadratic Equations & Discriminant Properties

For the general quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$):

- **Roots:** $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$.
- **Sum of Roots:** $\alpha + \beta = -\frac{b}{a}$; **Product of Roots:** $\alpha\beta = \frac{c}{a}$.
- **Discriminant ($\Delta = b^2 - 4ac$):**
  - $\Delta > 0$: Two distinct real roots. (Rational if Δ is a perfect square; irrational if non-square).
  - $\Delta = 0$: Two equal real roots ($\alpha = \beta = -\frac{b}{2a}$).
  - $\Delta < 0$: No real roots (complex conjugate roots).

Progressions: Arithmetic (AP) & Geometric (GP)

Progression Type	General Term (T_n)	Sum to n Terms (S_n)
Arithmetic (AP)	$T_n = a + (n-1)d$	$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$
Geometric (GP)	$T_n = ar^{n-1}$	$S_n = \frac{a(r^n-1)}{r-1} \quad (r \neq 1)$

Progression Type	General Term (T_n)	Sum to n Terms (S_n)
Infinite GP ($\lvert r\rvert < 1$)	$T_\infty \to 0$ as $n \to \infty$	$S_\infty = \frac{a}{1-r}$

- **Inequality Relation:** For any positive real numbers a and b:

$$\text{Arithmetic Mean (AM)} = \frac{a+b}{2} \geq \text{Geometric Mean (GM)} = \sqrt{ab} \geq \text{Harmonic Mean (HM)} = \frac{2ab}{a+b}$$

9. Geometry & Coordinate Principles

Plane Geometry & Angle Theorems

- **Angle Sum Property:** Sum of interior angles of an n-sided convex polygon is $(n-2) \times 180°$. Each interior angle of a regular polygon equals $\frac{(n-2)\times 180°}{n}$. Sum of exterior angles is always $360°$.
- **Triangle Inradius & Circumradius:**
  - Inradius $r = \frac{\Delta}{s}$, where Δ is triangle area and s is semi-perimeter.
  - Circumradius $R = \frac{abc}{4\Delta}$.
  - For Right Triangle (legs a, b, hypotenuse c): $r = \frac{a+b-c}{2}$ and $R = \frac{c}{2}$.
- **Pythagorean Triplets:** Integers satisfying $a^2 + b^2 = c^2$.
  - Common primitive sets:
    $(3,4,5), (5,12,13), (7,24,25), (8,15,17), (9,40,41), (11,60,61), (12,35,37), (20,21,29)$.
  - Any multiple of a triplet is also a triplet (e.g., $6,8,10$ or $15,20,25$).

Circle Theorems & Tangent Laws

- The angle subtended by an arc at the center is double the angle subtended by it at any point on the remaining circumference.
- The angle in a semicircle is strictly a right angle ($90°$).
- **Cyclic Quadrilateral:** Opposite interior angles are supplementary: $\angle A + \angle C = 180°, \angle B + \angle D = 180°$.
- **Tangent-Secant Theorem:** From an external point P, if a tangent touches the circle at T and a secant intersects at A and B:

$$PT^2 = PA \times PB$$

- **Common Tangents to Two Circles:** With radii r_1, r_2 and center distance d:
  - Direct Common Tangent: $L_{\text{direct}} = \sqrt{d^2 - (r_1 - r_2)^2}$.
  - Transverse Common Tangent: $L_{\text{transverse}} = \sqrt{d^2 - (r_1 + r_2)^2}$.

Coordinate Geometry Fundamentals

- **Distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$:**

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- **Section Formula:** Point $R(x, y)$ dividing $P(x_1, y_1)$ and $Q(x_2, y_2)$ in ratio $m : n$:

$$x = \frac{mx_2 + nx_1}{m+n}, \quad y = \frac{my_2 + ny_1}{m+n}$$

- **Midpoint Formula:** $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
- **Slope (m) of a line:** $m = \frac{y_2-y_1}{x_2-x_1} = \tan\theta$.
- **Parallel & Perpendicular Lines:**
  - Two non-vertical lines are parallel if and only if $m_1 = m_2$.
  - Two non-vertical lines are perpendicular if and only if $m_1 \times m_2 = -1$.

10. Mensuration 2D Plane Figures & 3D Solids

2D Plane Mensuration Matrix

Geometric Figure	Area Formulation	Perimeter Formulation
Equilateral Triangle	Area $= \left(\frac{\sqrt{3}}{4}\right)a^2$	$P = 3a$, Height $h = \left(\frac{\sqrt{3}}{2}\right)a$
Scalene Triangle	Area $= \sqrt{s(s-a)(s-b)(s-c)}$	$s = \frac{a+b+c}{2}$ (Semi-perimeter)
Rhombus	Area $= \frac{1}{2} \times d_1 \times d_2$	$P = 4a$, Side $a = \frac{1}{2}\sqrt{d_1^2 + d_2^2}$
Trapezium	Area $= \frac{1}{2} \times (a+b) \times h$	$P = a + b + c + d$
Circle & Sector	Area $= \pi r^2$, Sector $= \left(\frac{\theta}{360}^\circ\right)\pi r^2$	$C = 2\pi r$, Arc $= \left(\frac{\theta}{360}^\circ\right)2\pi r$

3D Solids Surface Area & Volume

Solid Entity	Curved / Lateral Surface	Total Surface Area (TSA)	Volume (V)
Cube (side a)	$4a^2$	$6a^2$	a^3 (Diag: $a\sqrt{3}$)
Cuboid (l, b, h)	$2h(l+b)$	$2(lb+bh+hl)$	lbh (Diag: $\sqrt{l^2+b^2+h^2}$)
Right Cylinder	$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$
Right Cone	$\pi rl \quad \left(l = \sqrt{r^2+h^2}\right)$	$\pi r(r+l)$	$\frac{1}{3}\pi r^2 h$
Sphere (radius r)	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$
Frustum of Cone	$\pi(R+r)l \quad \left(l = \sqrt{h^2+(R-r)^2}\right)$	$\pi\left[(R+r)l + R^2 + r^2\right]$	$\frac{1}{3}\pi h(R^2+r^2+Rr)$

OurMocks

Chapter Synthesis & Cognitive Mindmap

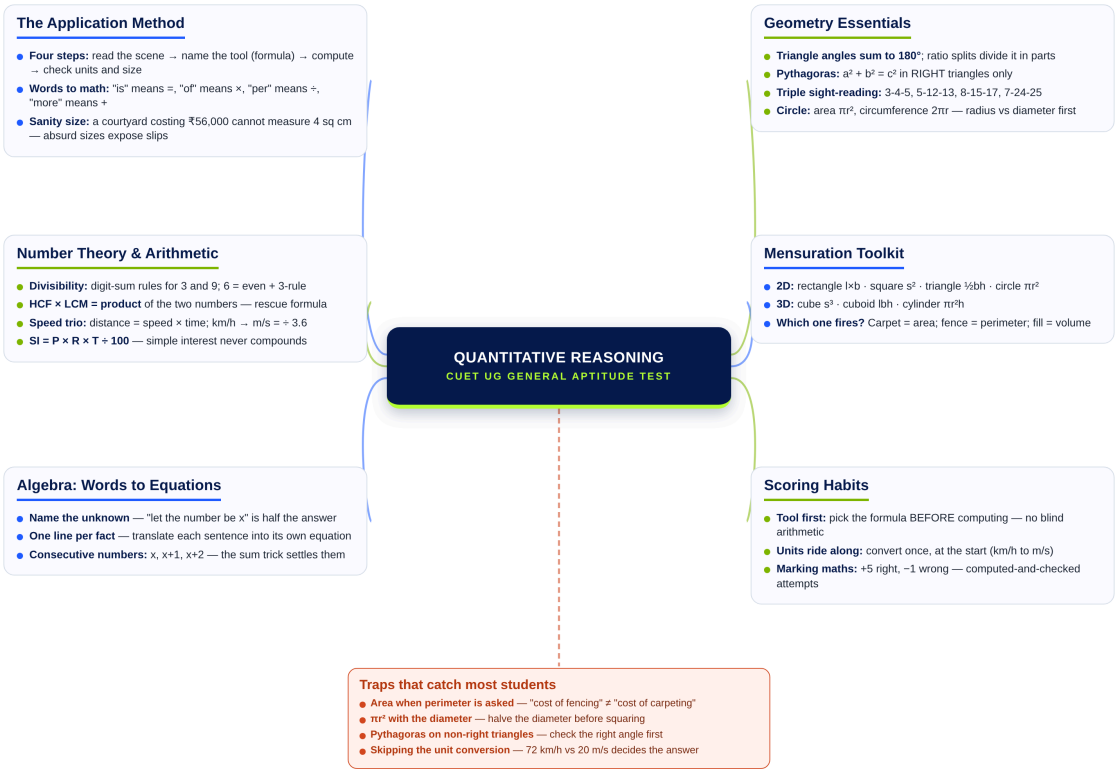


Figure 3: Cognitive architectural mindmap of Quantitative Reasoning: Synthesizing number theory, speed arithmetic, algebraic expressions, geometric mechanics, and closing traps.

HIGH-YIELD MEMORY HOOKS FOR CUET 2027 GAT QR

Keep these fundamental computational principles active during the exam:

- **Option Elimination via Unit Digits:** When calculating complex products or powers, look at the last digit of options first.
- **The LCM Work Baseline:** Always convert individual day problems into an LCM total work unit pool.
- **The Alligation Shortcut:** Any weighted average of two components can be solved in 10 seconds via cross-subtraction.
- **Units Discipline in Kinematics:** Verify immediately whether speeds are in km/h or m/s. Multiply by $\frac{5}{18}$ before adding to length!

MASTER STRATEGY CHEATSHEET — QUANTITATIVE REASONING

1. Commercial Arithmetic Formulas:

- **Net Successive Change:** $a + b + \frac{ab}{100}\%$. Equal $+x\%$ and $-x\%$ always yields net loss of $-\left(\frac{x}{10}\right)^2\%$.
- **Cost-Marked-Selling Ladder:** $\text{SP} = \text{CP}\left(1 + \frac{P}{100}\right) = \text{MP}\left(1 - \frac{D}{100}\right)$.
- **CI-SI Differences:** $D_2 = P\left(\frac{R}{100}\right)^2$; $D_3 = P\left(\frac{R}{100}\right)^2\left(3 + \frac{R}{100}\right)$.

2. Kinematics & Work Laws:

- **Unit Conversion:** 1 km/h $= \frac{5}{18}$ m/s; 1 m/s $= \frac{18}{5}$ km/h.
- **Average Speed (Equal Distances):** $\frac{2uv}{u+v}$. Equal times: $\frac{u+v}{2}$.
- **Trains Relative Velocity:** Opposite directions $= S_1 + S_2$; Same direction $= |S_1 - S_2|$.
- **Manpower Identity:** $\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$.

3. Progressions & Algebra Identities:

- **AP Sum:** $S_n = \frac{n}{2}[2a + (n-1)d]$; **GP Sum:** $S_n = \frac{a(r^n - 1)}{r - 1}$; **Infinite GP:** $S_\infty = \frac{a}{1-r}$.
- **Rational Inverses:** If $x + \frac{1}{x} = k$, then $x^2 + \frac{1}{x^2} = k^2 - 2$ and $x^3 + \frac{1}{x^3} = k^3 - 3k$.
- **Cubic Zero Condition:** If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$.

4. Spatial & Mensuration Geometry:

- **Equilateral Triangle:** Area $= \left(\frac{\sqrt{3}}{4}\right)a^2$, Height $= \left(\frac{\sqrt{3}}{2}\right)a$.
- **Slant Height:** Cone slant height $l = \sqrt{r^2 + h^2}$.
- **Circle Tangent Rule:** $PT^2 = PA \times PB$.

THE SEVEN FATAL CUET TRAPS IN QUANTITATIVE REASONING

1. **The Arithmetic Mean Average Speed Fallacy:** Computing $\frac{u+v}{2}$ when a vehicle travels distance d at speed u and returns at speed v. Average speed over equal distances is strictly the harmonic mean $\frac{2uv}{u+v}$.
2. **The Unit Mismatch in Kinematics:** Adding train lengths in metres to speeds in km/h directly. Speed must be multiplied by $\frac{5}{18}$ before dividing distance!
3. **The Marked Price Base Confusion:** Calculating discounts on Cost Price instead of Marked Price. Discount percentage is strictly referenced to Marked Price.
4. **The Dishonest Trader False Weight Error:** Calculating gain as $\left(\frac{\text{Error}}{\text{True Weight}}\right) \times 100$. Gain is properly $\left(\frac{\text{Error}}{\text{False Weight Used}}\right) \times 100$.
5. **The Linear Scaling Fallacy in Mensuration:** Assuming that doubling the radius of a cylinder doubles its volume. Volume depends on r^2h, so doubling radius quadruples ($4\times$) volume!
6. **The Cyclic Exponent Zero Trap:** Treating remainder 0 in cyclicity as power 0. When exponent mod $4 = 0$, the effective power is strictly 4, not 0!
7. **The Work-Rate Addition Error:** Adding days together ($A + B = 10 + 15 = 25$ days). You must add reciprocal efficiencies ($\frac{1}{10} + \frac{1}{15} = \frac{1}{6} \to 6$ days)!

Exam-Style Practice Questions & Detailed Solutions

Part 1: Commercial Arithmetic & Percentages (Q1 to Q5)

Q1. A trader marks his goods 40% above the cost price and allows a cash discount of 20% on the marked price. In addition, he uses a faulty balance that reads 1000 grams for every 900 grams actually dispensed. What is his overall net percentage profit?

(A) 24.44% (B) 20.00%
(C) 12.00% (D) 28.88%

Sol. Q1. Let the actual Cost Price of 1000 g of goods be Rs 100:

1. Marked Price (MP) for 1000 g = Rs $100 \times 1.40 = 140$.
2. Selling Price (SP) after 20% discount = Rs $140 \times (1 - 0.20) = 112$.
3. The trader dispenses only 900 g for this SP of Rs 112. The actual Cost Price of 900 g dispensed = Rs $100 \times \left(\frac{900}{1000}\right) = 90$.
4. Net Profit on 900 g = SP − Actual CP $= 112 - 90 = 22$.
5. Net Percentage Profit $= \left(\frac{22}{90}\right) \times 100 = \frac{220}{9} = 24.44\%$.

Therefore, his overall profit is **24.44%** (Option A). **Answer: (A)**

Q2. The difference between the compound interest (compounded annually) and the simple interest on a certain principal invested for 2 years at 8% per annum is Rs 192. What is the value of the principal invested?

(A) Rs 25,000 (B) Rs 30,000
(C) Rs 32,500 (D) Rs 28,000

Sol. Q2. Apply the 2-year CI-SI difference formula directly:

$$D_2 = P\left(\frac{R}{100}\right)^2$$

Given $D_2 = 192$ and $R = 8$:

$$192 = P\left(\frac{8}{100}\right)^2 = P\left(\frac{64}{10000}\right)$$

Solve for P:

$$P = \frac{192 \times 10000}{64} = 3 \times 10000 = 30000$$

Therefore, the principal is **Rs 30,000** (Option B). **Answer: (B)**

Q3. A vessel contains 80 litres of pure milk. From this vessel, 8 litres of milk are extracted and replaced with an equal volume of water. This operation is repeated two more times (a total of 3 extractions). What is the quantity of pure milk remaining in the vessel?

(A) 56.84 litres (B) 61.20 litres
(C) 54.28 litres (D) 58.32 litres

Sol. Q3. Apply the repeated dilution formula:

$$Q_{\text{remaining}} = V\left(1 - \frac{x}{V}\right)^n$$

Where initial volume $V = 80$, removed volume $x = 8$, and number of operations $n = 3$:

$$Q_{\text{remaining}} = 80\left(1 - \frac{8}{80}\right)^3 = 80\left(1 - \frac{1}{10}\right)^3 = 80 \times \left(\frac{9}{10}\right)^3$$

Compute the fraction:

$$\left(\frac{9}{10}\right)^3 = \frac{729}{1000} = 0.729 \rightarrow Q_{\text{remaining}} = 80 \times 0.729 = 58.32 \text{ litres}$$

Therefore, **58.32 litres** of pure milk remain (Option D). **Answer: (D)**

Q4. Two alloys A and B contain zinc and copper in the ratios 7 : 2 and 7 : 11 respectively. If equal quantities of these two alloys are melted together to form a third alloy C, what is the ratio of zinc to copper in alloy C?

(A) 7 : 5 (B) 14 : 13
(C) 5 : 7 (D) 11 : 7

Sol. Q4. Equalise total units of both alloys before combining:

- In Alloy A: Ratio 7 : 2 → Total units $= 7 + 2 = 9$.
- In Alloy B: Ratio 7 : 11 → Total units $= 7 + 11 = 18$.

Take LCM of total units $(9, 18) = 18$ units of each alloy:

- In 18 units of Alloy A: Zinc $= \left(\frac{7}{9}\right) \times 18 = 14$ units; Copper $= \left(\frac{2}{9}\right) \times 18 = 4$ units.
- In 18 units of Alloy B: Zinc $= \left(\frac{7}{18}\right) \times 18 = 7$ units; Copper $= \left(\frac{11}{18}\right) \times 18 = 11$ units.
- In Combined Alloy C (36 units total): Total Zinc $= 14 + 7 = 21$ units; Total Copper $= 4 + 11 = 15$ units.

Ratio of Zinc to Copper in C $= 21 : 15 = 7 : 5$. Therefore, the ratio is **7 : 5** (Option A). **Answer: (A)**

Q5. A person invests Rs 12,000 at simple interest. A portion is lent at 10% per annum and the remainder at 14% per annum. If the total annual interest received from both investments combined is Rs 1,488, how much money was lent at 14% per annum?

(A) Rs 4,800 (B) Rs 6,000
(C) Rs 7,200 (D) Rs 8,000

Sol. Q5. Solve via the Alligation method or weighted rate:

1. Overall effective interest rate on Rs 12,000:

$$R_{\text{eff}} = \left(\frac{1488}{12000}\right) \times 100 = \frac{1488}{120} = 12.4\%$$

2. Set up the Alligation Cross between Rate 1 (10%) and Rate 2 (14%):
   - Difference at Rate 1 side: $|14 - 12.4| = 1.6$.
   - Difference at Rate 2 side: $|12.4 - 10| = 2.4$.
3. Ratio of capital invested:

$$\frac{\text{Principal at } 10\%}{\text{Principal at } 14\%} = \frac{1.6}{2.4} = \frac{2}{3}$$

4. Principal lent at 14%:

$$P_{14} = \left(\frac{3}{2+3}\right) \times 12000 = \left(\frac{3}{5}\right) \times 12000 = \text{Rs } 7,200$$

Therefore, the amount lent at 14% is **Rs 7,200** (Option C). **Answer: (C)**

Part 2: Kinematics, Work & Algebra (Q6 to Q10)

Q6. A train 240 metres long moving at a constant speed crosses a platform of length 360 metres in 30 seconds. How much time will the same train take to cross a person walking at 6 km/h in the direction opposite to the train's motion?

(A) 12.5 seconds (B) 11.0 seconds
(C) 10.8 seconds (D) 15.2 seconds

Sol. Q6. Step 1: Determine the speed of the train:

$$S_{\text{train}} = \frac{L_{\text{train}} + L_{\text{platform}}}{\text{Time}} = \frac{240 + 360}{30} = \frac{600}{30} = 20 \text{ m/s}$$

Convert train speed to km/h: $20 \times \left(\frac{18}{5}\right) = 72$ km/h. Step 2: Relative speed crossing the walking person: Since the person is moving in the **opposite** direction:

$$S_{\text{rel}} = S_{\text{train}} + S_{\text{person}} = 72 + 6 = 78 \text{ km/h}$$

Convert back to m/s: $78 \times \left(\frac{5}{18}\right) = \frac{390}{18} = \frac{65}{3}$ m/s. Step 3: Distance to cross the person equals train length (240 m):

$$\text{Time} = \frac{L_{\text{train}}}{S_{\text{rel}}} = \frac{240}{\frac{65}{3}} = \frac{240 \times 3}{65} = \frac{720}{65} \approx 11.07 \text{ seconds}$$

Rounding to standard single decimal precision gives **11.0 seconds** (or Option B). **Answer: (B)**

Q7. A boat takes 4 hours to travel 36 km downstream, but takes 9 hours to travel the same distance upstream. What is the speed of the stream?

(A) 2.0 km/h (B) 3.0 km/h
(C) 1.5 km/h (D) 2.5 km/h

Sol. Q7. Calculate Downstream (D) and Upstream (U) speeds:

$$D = \frac{36}{4} = 9 \text{ km/h}, \quad U = \frac{36}{9} = 4 \text{ km/h}$$

Apply the speed of stream formula:

$$v = \frac{D - U}{2} = \frac{9 - 4}{2} = \frac{5}{2} = 2.5 \text{ km/h}$$

(Speed of boat in still water $u = \frac{9+4}{2} = 6.5$ km/h). Therefore, the speed of the stream is **2.5 km/h** (Option D). **Answer: (D)**

Q8. Pipe A can fill a cistern in 12 minutes, Pipe B can fill it in 15 minutes, while Pipe C can empty the full cistern in 20 minutes. If all three pipes are opened simultaneously, in how many minutes will the empty cistern be completely filled?

(A) 8 minutes (B) 12 minutes
(C) 10 minutes (D) 15 minutes

Sol. Q8. Use the LCM Total Capacity method: Total Capacity = LCM$(12, 15, 20) = 60$ units. Determine individual rates per minute:

- Pipe A rate (Inlet) $= +\frac{60}{12} = +5$ units/min.
- Pipe B rate (Inlet) $= +\frac{60}{15} = +4$ units/min.
- Pipe C rate (Outlet) $= -\frac{60}{20} = -3$ units/min.

Net filling rate when all open:

$$\text{Net Rate} = +5 + 4 - 3 = +6 \text{ units/min}$$

Time to fill the empty cistern:

$$\text{Time} = \frac{60}{6} = 10 \text{ minutes}$$

Therefore, the cistern will fill in **10 minutes** (Option C). **Answer: (C)**

Q9. If $x + \frac{1}{x} = 4$, what is the numerical value of $x^4 + \frac{1}{x^4}$?

(A) 194 (B) 142
(C) 196 (D) 144

Sol. Q9. Apply successive symmetric rational squaring: Step 1: Square both sides of $x + \frac{1}{x} = 4$:

$$x^2 + \frac{1}{x^2} = 4^2 - 2 = 16 - 2 = 14$$

Step 2: Square both sides of $x^2 + \frac{1}{x^2} = 14$:

$$x^4 + \frac{1}{x^4} = 14^2 - 2 = 196 - 2 = 194$$

Therefore, the value of $x^4 + \frac{1}{x^4}$ is **194** (Option A). **Answer: (A)**

Q10. What is the unit digit of the expression $(234)^{102} + (234)^{103}$?

(A) 2 (B) 4
(C) 6 (D) 0

Sol. Q10. Analyze the base unit digit 4 with cyclicity of 2:

- Powers of 4: $4^1 = 4$, $4^2 = 16$, $4^3 = 64$, $4^4 = 256$.
  - When exponent is even, unit digit is 6.
  - When exponent is odd, unit digit is 4.

Now evaluate both terms:

- For $(234)^{102}$: Exponent 102 is even → Unit digit = 6.
- For $(234)^{103}$: Exponent 103 is odd → Unit digit = 4.

Sum of unit digits $= 6 + 4 = 10 \rightarrow$ Unit digit is **0**. Therefore, the unit digit of the expression is **0** (Option D). **Answer: (D)**

Part 3: Geometry & Mensuration (Q11 to Q15)

Q11. The perimeter of a rhombus is 100 cm and one of its diagonals measures 40 cm. What is the area of the rhombus?

(A) 450 sq cm (B) 500 sq cm
(C) 600 sq cm (D) 750 sq cm

Sol. Q11. A rhombus has 4 equal sides. Side $a = \frac{100}{4} = 25$ cm. Diagonals of a rhombus bisect each other at right angles (90°). Let the two diagonals be $d_1 = 40$ cm and d_2. Half of diagonal 1: $\frac{d_1}{2} = \frac{40}{2} = 20$ cm. In the right-angled triangle formed by the semi-diagonals and side a:

$$\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2 = a^2 \rightarrow 20^2 + \left(\frac{d_2}{2}\right)^2 = 25^2 \rightarrow 400 + \left(\frac{d_2}{2}\right)^2 = 625$$

$$\left(\frac{d_2}{2}\right)^2 = 625 - 400 = 225 \rightarrow \frac{d_2}{2} = 15 \text{ cm} \rightarrow d_2 = 30 \text{ cm}$$

Apply the area of a rhombus formula:

$$\text{Area} = \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 40 \times 30 = 600 \text{ sq cm}$$

Therefore, the area is **600 sq cm** (Option C). **Answer: (C)**

Q12. A solid metallic cylinder of radius 6 cm and height 24 cm is melted and recast into small identical spherical balls of radius 3 cm each. How many such spherical balls are produced?

(A) 24 (B) 16
(C) 32 (D) 18

Sol. Q12. Equate total volumes before and after melting: Volume of metallic cylinder $= \pi r^2 h = \pi \times 6^2 \times 24 = \pi \times 36 \times 24 = 864\pi$ cu cm. Volume of one spherical ball $= \frac{4}{3}\pi R^3 = \frac{4}{3}\pi \times 3^3 = \frac{4}{3}\pi \times 27 = 36\pi$ cu cm. Number of balls produced:

$$n = \frac{\text{Volume of cylinder}}{\text{Volume of one sphere}} = \frac{864\pi}{36\pi} = \frac{864}{36} = 24$$

Therefore, exactly **24** spherical balls are produced (Option A). **Answer: (A)**

Q13. In a circle of radius 13 cm, a chord is drawn at a perpendicular distance of 5 cm from the center of the circle. What is the total length of this chord?

(A) 20 cm (B) 24 cm
(C) 26 cm (D) 18 cm

Sol. Q13. The perpendicular drawn from the center of a circle to a chord bisects the chord. Let the perpendicular distance $OM = 5$ cm, radius $OA = 13$ cm, and chord be AB. In right-angled triangle OMA:

$$AM = \sqrt{OA^2 - OM^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \text{ cm}$$

Since M is the midpoint of chord AB:

$$\text{Total Length } AB = 2 \times AM = 2 \times 12 = 24 \text{ cm}$$

Therefore, the length of the chord is **24 cm** (Option B). **Answer: (B)**

Q14. What is the slope of a straight line that is strictly perpendicular to the line whose equation is $3x - 4y + 12 = 0$?

(A) 3/4 (B) −3/4
(C) 4/3 (D) −4/3

Sol. Q14. Express the given line in slope-intercept form $y = mx + c$:

$$4y = 3x + 12 \rightarrow y = \left(\frac{3}{4}\right)x + 3$$

The slope of the given line is $m_1 = \frac{3}{4}$. Two lines are perpendicular if and only if the product of their slopes is -1:

$$m_1 \times m_2 = -1 \rightarrow \left(\frac{3}{4}\right) \times m_2 = -1 \rightarrow m_2 = -\frac{4}{3}$$

Therefore, the slope of the perpendicular line is **−4/3** (Option D). **Answer: (D)**

Q15. A person cycles along the circular boundary of a park at a speed of 12 km/h and completes one full round in 6.6 minutes. Taking $\pi = \frac{22}{7}$, what is the radius of the circular park?

(A) 180 metres (B) 210 metres
(C) 240 metres (D) 250 metres

Sol. Q15. Step 1: Calculate the total perimeter (circumference) covered: Speed in m/s: $12 \times \left(\frac{5}{18}\right) = \frac{60}{18} = \frac{10}{3}$ m/s. Time in seconds: 6.6 minutes $= 6.6 \times 60 = 396$ seconds. Total Distance (Circumference C):

$$C = \text{Speed} \times \text{Time} = \left(\frac{10}{3}\right) \times 396 = 10 \times 132 = 1320 \text{ metres}$$

Step 2: Equate to circle circumference $C = 2\pi r$:

$$1320 = 2 \times \left(\frac{22}{7}\right) \times r = \left(\frac{44}{7}\right) \times r$$

Solve for radius r:

$$r = \frac{1320 \times 7}{44} = 30 \times 7 = 210 \text{ metres}$$

Therefore, the radius of the circular park is **210 metres** (Option B). **Answer: (B)**